

Mr. Surya B K Singh

FUNCTIONAL HEAD: PROCESS DEVELOPMENT/TECHNOLOGY TRANSFER LEAD

PERSONAL PROFILE:

DATE OF BIRTH: 15/06/1980

NATIONALITY: INDIAN

LANGUAGE: ENGLISH, HINDI & GUJARATI



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MS BIOTECHNOLOGY: BANARAS HINDU UNIVERSITY (BHU) YEAR: 2004-2006

PURSuing PHD: DEPARTMENT OF BIOTECHNOLOGY at GURU GHASIDAS VISHWAVIDYALAYA, BILASPUR

PROFESSIONAL PROFILE

20-year experience in Biopharmaceutical industry hands on and Managerial experience in handling process development and Technology Transfer/MSAT of recombinant therapeutic products. Collaboration, planning, presentation and troubleshooting are key skills in managing the activities. Handled products like high value recombinant peptides in both macular degeneration segment/diabetic/osteoporosis/Cancer from concept to manufacturing stage I was fortunate to work both on innovative and biosimilar proteins and these have helped me in fast-tracking the R&D development of any kind of proteins thus saving time for clinical and production activities. Hands on Process/impurity characterization/ bio-similarity to meet EMA /US-FDA requirement, in-process method development, sample analysis, Pilot scale process transfer, optimization, operations and write-up, review of tech transfer documentation are my key strengths.

PATENTS/ PAPER PUBLICATIONS AND PRESENTATIONS:

- **Patents:**
 - ✓ METHOF FOR PURIFICATION OF HYALURONIDASE ENZYME. PCT/IB2025/056717, IN202421053456
 - ✓ DEVELOPMENT OF NOVEL FORMULATION FOR STABILIZATION RECOMBINANT HUMAN HYALURONIDASE (rHuPH20), E-1/31022/2025-MUM
 - ✓ METHOD FOR PURIFICATION OF GONADOTROPIN HORMONE, E-I/60126/2024-MUM.
 - ✓ AN IMPROVE PROCESS FOR THE PREPARATION OF PHARMACEUTICALS GRADE INTERFERON ALPHA 2B. b. WO 2016/079598 A1
 - ✓ AN IMPROVE PROCESS FOR THE PURIFICATION OF INTERFERON ALPHA 2B EXPRESSED IN SOLUBLE E. COLI. Application No: 3628/MUM/2014A
 - ✓ Ranibizumab novel refolding process (Under review)
- **Published papers:**
 - ✓ Alternative Conformation Model of seed protein *Dek1* for better understanding of Structure-Function Relationship, Journal of Protein and Proteomics, Vol. 1, No. 2, July-December 2010, pp. 77-90.
 - ✓ Purification of Tagged protein over produced in *E. coli*, May. 20, 2012, *CURRENT SCIENCE*.

FUNCTIONAL HEAD- INTAS PHARMACEUTICALS LIMITED

INTAS Pharmaceuticals Limited, Biotechnology R&D: May 2023 to Present (3 Year)

Products: Enzyme (Hyaluronidase), ADC (TDM 1 & Enhertu), MAB (Nivolumab & Pembrolizumab), LH, HCG, Ant-D.

- ✓ Vertical Lead downstream process development / Scale up of ADC, Mammalian/Hormones/Enzymes and Microbial biosimilar Platform
- ✓ To Conduct Technology transfer.
- ✓ To Enable Scaleup for commercial manufacturing
- ✓ Four to Five Managers reporting for 12 different projects and total 25 people team size.
- ✓ Process Characterization and control strategies.
- ✓ Manage Scientist (goal setting, understanding development need, assigning responsibility in lab/managerial activity, critical assessing their performance and provide timely feedback)

- ✓ To strengthen the team to develop understanding of dsp development, consistency batches, engineering campaign, CT campaign, generation of DS for pre-clinical and clinical studies., virus validation, process improvement, process Characterization by QbD approach, resin/TFF reuse studies.
- ✓ Work collaboration with DPD, USP, MSAT and BBM of completion of tech transfer.
- ✓ Author and review of TTD/CTD section for regulatory submission.
- ✓ Knowledge of large-scale chromatography, TFF operation and data analysis. To face FDA audit for current organization, key people from downstream side.
- ✓ To remove HCP from process and check bioactivity for different hydrophobic variant in Aflibercept and impact in drug substance.

GROUP LEAD- LUPIN

Lupin Limited, Biotechnology R&D: August 2019 to May 2023 (4 Year)

Products: Ranibizumab, Aflibercept, Nivolumab and Ustekinumab (Fc, Fab and Fc+Fab).

- ✓ Evolution of vendor capability, Equipment qualification & resin life cycle/Cost effective and Regulatory document like IMPD and CMC for EU/US.
- ✓ Membrane/resin life cycle and reusability studies without effecting CQA.
- ✓ To write pre PPQ-Technology transfer document for US/EU, process development summary, Scale up reports, review FMEA, PAR, NOR and MOR document, Control strategy document, Comparability reports, impurity assessment, impurity spike studies and DOE for Ranibizumab and Aflibercept.
- ✓ Impurities Isolation (RP-HPLC/CEX/SEC/HIC)/clearance spike studies and impurity characterization, product characterization, media optimization.
- ✓ Scale up and scale down operation qualification, Improvement of refolding process in Ranibizumab.
- ✓ To increase productivity more than 3X of DS by using robust chromatography steps and refolding of Ranibizumab.
- ✓ Knowledge of large-scale chromatography, TFF operation and data analysis. To face FDA audit for current/past organization, key people from downstream department.
- ✓ To remove HcP from process and check bioactivity for different hydrophobic variant in Aflibercept and impact in drug substance.
- ✓ To develop robust process, highest purity level comparable to competitor for Aflibercept and Ranibizumab.
- ✓ Product characterization with the help of DOE-JMP/design expert (Plackett–Burman design/Multivariant/full factorials, OFAT) and reports preparation.
- ✓ People Interview, Review document like PDR, SDM qualifications, Scale-up, Impurity reports, Risk assessment protocol/reports, Process Characterizations protocol/reports, BPR, SPRs, MFR, CQA protocol/reports, Instrument IQ, OQ, PQ, URS, Change control, deviation & investigation and People Management – team of 8 people (M Sc./M Tec/PhD)

PREVIOUS WORK EXPERIENCE

YEARS	ORGANISATION	PRODUCTS	EXPERIENCE
AUG'19- MAY'23	LUPIN LIMITED	Ranibizumab, Aflibercept, Ustekinumab	Technology transfer and resin life cycle/Cost effective. People management – team of 15 people
OCT'14- JUL'19	STELIS BIOPHARMA (Stride Arcolab Limited)	GLARGINE LISPRO PTH Beva	Process design techniques like DoE, Product maximization - quantity & quality devt. Report evolution to pilot and production compliant documentation and process transfer. People management – team of 10 people
NOV'09- SEPT'14	UNICHEM Laboratories Limited	Human INSULIN GLARGINE IFN α 2B	Design, development, scale-up, tech transfer from R&D→Pilot Patent writing, advance process devt. Product life cycle.

			People management – team of 6 people
JUL'07- OCT'09	TARGET-IN - CENTURY	FcE fusion protein	Novel protein expression, purification by IEX, HIC, formulation development.
JAN'06- JUN'07	INTAS Pharmaceuticals Limited	GCSF	Downstream operations, chromatography system, western Blot and document writing, Refolding of protein.
JAN'06	BHU	Dek 1	Purification and understanding of basic downstream instrument, Purification of protein by GFC. SDS-PAGE analysis, Literature search, Instrumentation.

EQUIPMENT & PROCESS EXPERTISE

Process area	PROCESS SCALE, LITERS	EQUIPMENT
Downstream – R&D	0.4L to 10L	AKTA explorer, AKTA basic, AKTA prime, Bio-Rad purification system, TFF system (50cm ² to 0.5 m ²), Viral filtration, Refolding vessels, Hollow fibre filtration,
Downstream – Pilot	50L to 200L	AKTA pilot, RP-HPLC system, Millipore TFF 4 m ² system, AKTA Process, TFF system (0.5 m ² to 2m ²), Column (up to 200mm)
Downstream – Production	Microbial: 100L to 1KL Mammalians: 50L to 2KL	Homogenizer: GEA 3006 (100L/h) Centrifuge: Westphalia HFDF/MFDF/DFF: pall life science CF 5L/min/m ² Chromatography: AKTA process, AKTA Pilot Column: 200/300/500 mm Crystallization, Precipitation Lyophilisation: LyoStar Formulation Refolding Vessels: 5KL vessels
In-process/ Analytical	Downstream	SDS- PAGE: Bio-Rad, Coomassie, Silver staining Western Blot: Bio-Rad RPC/SEC/IE HPLC: Waters, Agilent UV-flour spec : Thermo GE,
Upstream – R&D	0.4L	CUBOTA Centrifuge, Sartorius Fermenter, GEA high pressure homogeniser.

REFERENCE

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